

Mubashir Akhtar (Metallurgy and Materials Engineer)

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ABOUT ME

Metallurgy and Materials Engineer with a strong academic background experience in materials processing, nanotechnology, and industrial research. Skilled in applying advanced techniques to enhance performance, sustainability, and efficiency in manufacturing environments. Passionate about driving innovation and contributing to high-impact engineering solutions.

EDUCATION

Bachelor of Science in Metallurgy and Materials Engineering.

Pakistan Institute of Engineering and Applied Sciences (PIEAS) Islamabad, Pakistan. [12/09/2021 – 01/07/2025]

Website: <https://www.pieas.edu.pk/>

Final grade: 3.29/4.00 **Level in EQF:** EQF level 6

Core subjects: Thermodynamics, Physical metallurgy, Corrosion, Heat Treatment, Ferrous metallurgy, Melting and casting, Phase Transformation, Polymers, Functional Materials, Advanced Materials and Vacuum Technology.

SKILLS

- **Soft skills:** Communication, teamwork, problem-solving, adaptability, time management, leadership, interpersonal skills, conflict resolution
 - **Standards & Testing:** Non-destructive testing (NDT), Characterization, Heat Treatment, Tensile Testing, Hardness Testing, Metallography, Casting, Casting Simulation, Corrosion Protection, XRD, SEM, TGA, FTIR, Additive Manufacturing
 - **Software & Simulation:** ProCAST, HEAP, Origin, Microsoft Office, SolidWorks, Xpert High Score, ImageJ and Autodesk.
 - **Programming:** C++, Python, MATLAB.
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RESEARCH WORK (THESIS)

- **Synthesis and Enhancement of antimicrobial and catalytic properties of Quasicrystals based on Al-Cu-Fe System by chromium doping**
Quasicrystals with antimicrobial properties and corrosion resistance.
Catalytic behavior for research and industrial use.
 - **Synthesis of a Catalytic Material for Environmental Degradation of Polyethylene.**
Novel material capable of degrading polyethylene waste.
Sustainable pathway for mitigating long-term environmental impact
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INTERNSHIP EXPERIENCE

- **Research Assistant NRP Funded Project**
Department of Metallurgy and Materials Engineering, PIEAS
Project Title: Synthesis and Enhancement of Antimicrobial and Catalytic Behavior of Quasicrystals
Duration: June 2024 – June 2025
- **National Centre for Physics (NCP) Islamabad.**
Find inconsistencies in 3D printing by using ISO guidelines.
Optimized print parameters and quality checks.
Improve strength, accuracy, and process efficiency.
- **National Institute of Lasers and Optronics Islamabad.**
Applied lithium coating on tungsten substrate for nuclear reactor applications.
Enhance performance to improve thermal stability, reduce tritium retention, and increase surface compatibility.

PROJECTS

1. Microstructural Evolution and Property Enhancement of Aluminum through Strontium.
 2. Industrial Project on Optimization of gating design, using simulation, to produce defect free casting.
 3. Synthesis of MXene from MAX Phase, Enabling Advanced Applications in Batteries, Supercapacitors, and Electromagnetic Protection.
 4. Advanced Exfoliation Techniques for Graphene Synthesis and Applications.
 5. Development of Graphene-Enhanced PVDF Composite Sheets for Advanced EMI Shielding Applications.
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CONFERENCES AND SEMINARS.

- **International Workshop on Nanomaterials and Catalysis IWNMC 23.**
Location: Pakistan Institute of Engineering and Applied Sciences (PIEAS), Islamabad.
 - **Effect of Microstructure on the Localized Corrosion and Atmospheric Stress Corrosion Cracking of 15-5PH Stainless Steel (Joint event with ICORR).**
Location: North Scottish Branch and the Institute of Corrosion Online.
 - **International conference on innovations in biomaterials and nanotechnology (BioNanoCon24).**
Location: Institute of Space and Technology, Islamabad.
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CERTIFICATES AND COURSES.

- **Non-Destructive Testing (NDT) for Engineer.**
National Centre for Non-Destructive testing Islamabad.
 - **Materials Data Science and Informatics by Coursera.**
Georgia Institute of Technology USA.
 - **Certificate of participation in an online Course 'AI Materials'.**
Korean Advance Institute of Science and Technology (KAIST), South Korea.
 - **Online Course on "Nanotechnology: A Markers Course".**
Duke University and North Carolina State University (USA).
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CO-CURRICULAR ACTIVITIES.

- Head of PIEAS wing at University Hub Pakistan (UHP).
- TEDx PIEAS Management Team Head.
- Management Head at VAS PIEAS.
- Member of Cricket Team of University